

Polyamide-Imide (PAI)

PAI is a thermoplastic, further classified an an imide plastic. It can have a moderately high tensile strength among the thermoplastics in the database.

The properties of PAI include four common variations. This page shows summary ranges across all of them. For more specific values, follow the links immediately below. The graph bars on the material properties cards further below compare PAI to: imide plastics (top), all thermoplastics (middle), and the entire database (bottom). A full bar means this is the highest value in the relevant set. A half-full bar means it's 50% of the highest, and so on.

- 30% Carbon Fiber (30 CF) PAI
- 30% Glass Fiber (30 GF) PAI
- Low Friction PAI
- Unfilled PAI

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Mechanical Properties

Compressive (Crushing) Strength

260 to 300 MPa

37 to 44 x 10³ psi

Elastic (Young's, Tensile) Modulus

4.0 to 8.3 GPa

0.58 to 1.2 x 10⁶ psi

Elongation at Break

3.0 to 12 %

Flexural Modulus

4.0 to 6.8 GPa

0.58 to 1.0 x 10⁶ psi

Flexural Strength

160 to 320 MPa

23 to 46 x 10³ psi

Impact Strength: Notched Izod

45 to 140 J/m

0.84 to 2.5 ft-lb/in

Tensile Strength: Ultimate (UTS)

100 to 180 MPa

15 to 26 x 10³ psi

Thermal Properties

Glass Transition Temperature

280 °C

530 °F

Heat Deflection Temperature At 1.82 MPa (264 psi)

280 °C

530 to 540 °F

Maximum Temperature: Autoignition

530 °C

980 °F

Maximum Temperature: Decomposition

490 °C

910 °F

Maximum Temperature: Mechanical

260 °C

500 °F

Specific Heat Capacity

910 to 1000 J/kg-K

0.22 to 0.24 BTU/lb-°F

Thermal Conductivity

0.26 to 0.53 W/m-K

0.15 to 0.31 BTU/h-ft-°F

Thermal Expansion

15 to 34 µm/m-K

Otherwise Unclassified Properties

Density

1.4 to 1.6 g/cm³

87 to 99 lb/ft³

Limiting Oxygen Index (LOI)

45 %

Water Absorption After 24 Hours

0.3 to 0.4 %

Water Absorption At Saturation

1.5 to 1.7 %

Common Calculations

Stiffness to Weight: Axial

1.6 to 3.2 points

Stiffness to Weight: Bending

38 to 47 points

Strength to Weight: Axial

19 to 34 points

Strength to Weight: Bending

33 to 49 points

Thermal Diffusivity

0.16 to 0.41 mm²/s

Thermal Shock Resistance

48 to 110 points

Followup Questions

How are the material properties defined?

How does PAI compare to other thermoplastics?

How does it compare to other polymeric materials?

Further Reading

Reinforced Plastics Durability, Geoffrey Pritchard (editor), 1999

SPI Plastics Engineering Handbook of the Society of the Plastics Industry, Inc., 5th ed., Michael L. Berins (editor), 2000

Modern Plastics Handbook, Charles A. Harper (editor), 1999

Plastics Materials, 7th ed., J. A. Brydson, 1999

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